

What is this?

This notebook is part of a collection of `pLuto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the Università dell'Insubria in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.

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TIME-DOMAIN ASTROPHYSICS

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Singular-Spectrum Analysis

- We want to introduce the **singular-spectrum analysis (SSA)**. In simple words, SSA decomposes a time series into a set of summable components that are grouped together and interpreted as *trend*, *periodicity* and *noise*.
- SSA emphasises **separability** of the underlying components, and can readily separate periodicities that occur on different time scales, even in very noisy time series data. The original time series is recovered by summing together all of its components.
- SSA can be used to analyse and reconstruct a time series with or without different components as desired. For example, you could apply SSA to:
 1. construct a smoothed version of a time series using a small subset of its components;
 2. investigate a time series' periodic components to understand the underlying processes that generated the time series;
 3. reconstruct the original time series without its periodic components;
 4. remove all trend and periodic components from the series, leaving just the 'noise', which may be meaningful in and of itself...
- Unlike the commonly used autoregressive integrated moving average (ARIMA) method, SSA makes no assumptions about the nature of the time series, and has just a single adjustable (and easily interpretable) parameter.
- Singular-Spectrum Analysis is connected to the popular exploratory and dimensionality reduction technique Principal Component Analysis. An introduction to the PCA can be found here ([notebook](#), [html](#)).

A Toy Time Series

.....

- We'll first define an arbitrary, toy time series, $F = \{f_0, f_1, \dots, f_{N-1}\}$, containing trend, periodic and noise components:

$$f_t = 0.001 \times (t - 100)^2 + 2 \sin\left(\frac{2\pi t}{p_1}\right) + 0.75 \sin\left(\frac{2\pi t}{p_2}\right) + \text{Rand}\{-1, 1\}$$

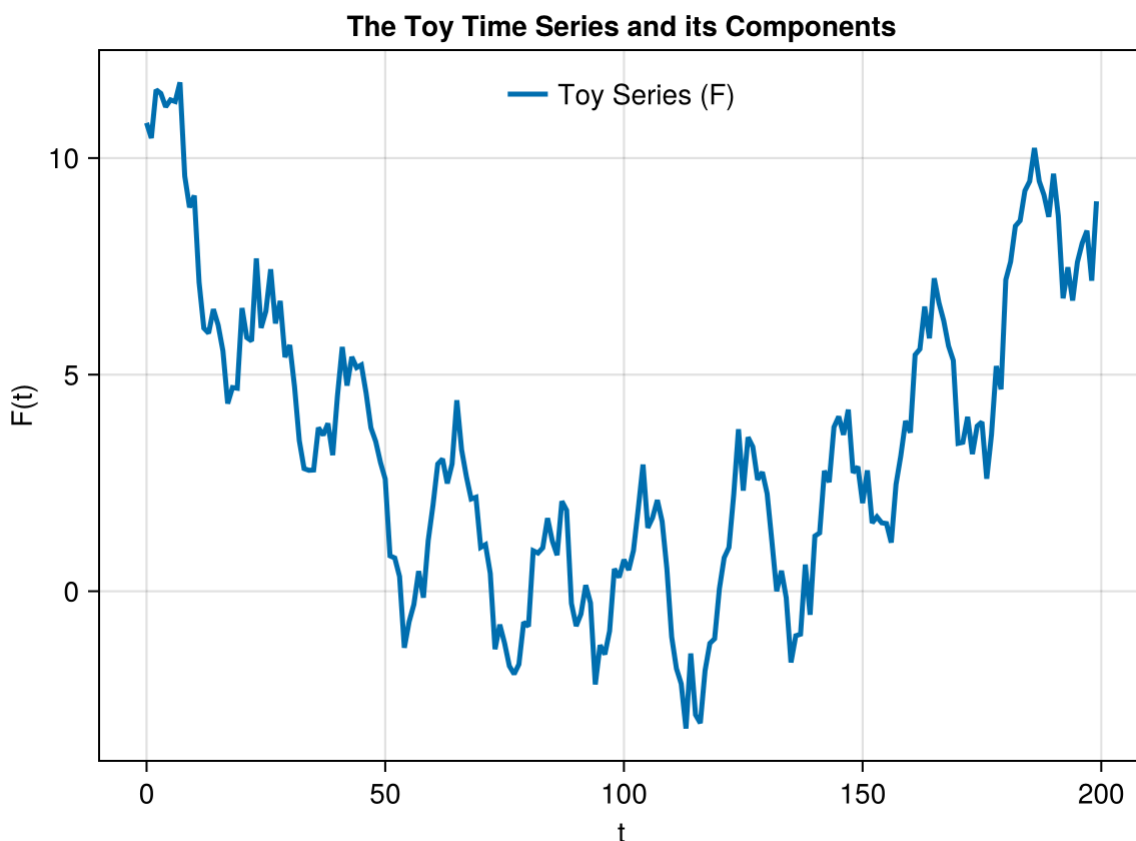
- where $t = \{0, 1, \dots, N - 1\}$ is a discrete time moment, p_1 and p_2 are set time periods and $\text{Rand}\{-1, 1\}$ is a random number uniformly distributed between -1 and 1 .
- The $0.001 \times (t - 100)^2$ term defines the (parabolic) trend of the series, $2 \sin(\frac{2\pi t}{p_1})$ and $0.75 \sin(\frac{2\pi t}{p_2})$ are two periodic components with differing periodicities and amplitudes.

```

1 begin
2   N = 200
3   t = 0:N-1
4   trend = 0.001 * (t - 100).^2
5   p1, p2 = 20, 30
6   periodic1 = 2 * sin.(2*pi*t/p1)
7   periodic2 = 0.75 * sin.(2*pi*t/p2)
8
9   Random.seed!(123)
10  noise = 2 .* (rand(N) .- 0.5)
11  F = trend .+ periodic1 .+ periodic2 .+ noise
12 end;

```

Total time-series only? ☒



- Based on visual inspection, the presence of trend (yellow), periodic (green and magenta) and noise (cyan) components in the time series is clear. However, the second periodic component (magenta) in the toy series is not apparent. Could we use SSA to recover the trend, periodic and noise components from the toy time series?

Introducing the SSA Method

From a Time Series to a Trajectory Matrix

- The first step of SSA is to map the time series F to a sequence of multi-dimensional lagged vectors.
- Let an integer L be the *window length*, $2 \leq L \leq N/2$. We form a ‘window’, given by the subseries $\{f_i, f_{i+1}, \dots, f_{i+L-1}\}$, for $i = 0, \dots, N - L$.
- We slide this window along the time series, forming a column vector, $\mathbf{X}_i$, for each subseries. That is, we have:

$$\begin{aligned}
 \mathbf{X}_0 &= (f_0, f_1, f_2, \dots, f_{L-1})^T \\
 \mathbf{X}_1 &= (f_1, f_2, f_3, \dots, f_L)^T \\
 \mathbf{X}_2 &= (f_2, f_3, f_4, \dots, f_{L+1})^T \\
 \mathbf{X}_3 &= (f_3, f_4, f_5, \dots, f_{L+2})^T \\
 &\vdots \\
 \mathbf{X}_{N-L} &= (f_{N-L}, f_{N-L+1}, f_{N-L+2}, \dots, f_{N-1})^T.
 \end{aligned}$$

- These column vectors form the L -trajectory matrix, $\mathbf{X}$, of the time series (hereafter just *trajectory matrix*):

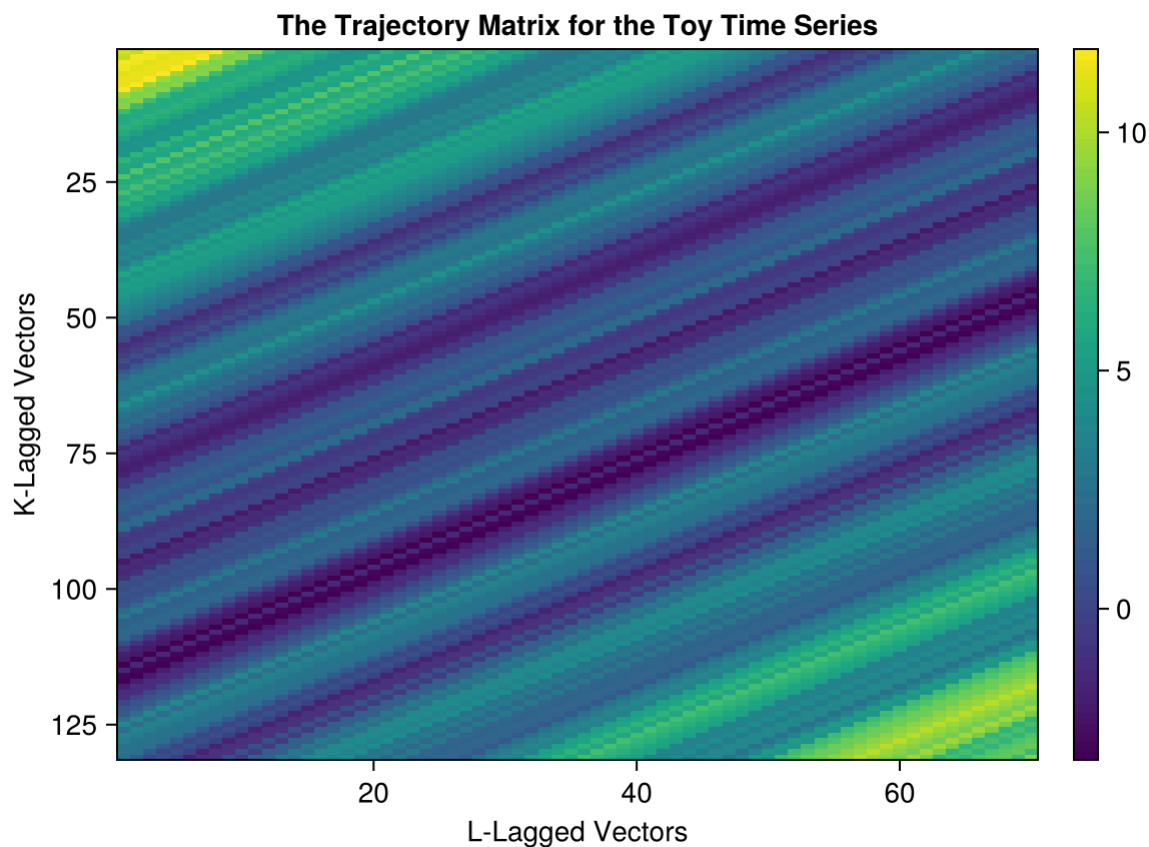
$$\mathbf{X} = \begin{bmatrix} f_0 & f_1 & f_2 & f_3 & \dots & f_{N-L} \\ f_1 & f_2 & f_3 & f_4 & \dots & f_{N-L+1} \\ f_2 & f_3 & f_4 & f_5 & \dots & f_{N-L+2} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ f_{L-1} & f_L & f_{L+1} & f_{L+2} & \dots & f_{N-1} \end{bmatrix}$$

- From writing out the matrix above, it is clear that the elements of the *anti-diagonals* (that is, the diagonals running from bottom-left to top-right) are equal. This type of matrix is known as a Hankel matrix.
- For our toy time series, we'll set the window length to 70, and defer discussion on how to select an appropriate window length later.
- Let $K = N - L + 1$ represent the number of columns in the trajectory matrix. We'll refer to the columns of $\mathbf{X}$ as the L -lagged vectors, and the rows as K -lagged vectors.

```

1 begin
2   L = 70 # The window length.
3   K = N - L + 1 # The number of columns in the trajectory matrix.
4   # Create the trajectory matrix by pulling the relevant subseries of F, and
   # stacking them as columns.
5   as = [F[i:i+L-1] for i in range(1,K)]
6   X = reduce(hcat, as)
7 end;

```



Decomposing the Trajectory Matrix

- The second step is decomposing the trajectory matrix with a singular-value decomposition (SVD).

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

- where:
 - $\mathbf{U}$ is an $L \times L$ unitary matrix containing the orthonormal set of *left singular vectors* of $\mathbf{X}$ as columns;
 - $\mathbf{\Sigma}$ is an $L \times K$ rectangular diagonal matrix containing L *singular values* of $\mathbf{X}$, ordered from largest to smallest; and
 - $\mathbf{V}$ is a $K \times K$ unitary matrix containing the orthonormal set of *right singular vectors* of $\mathbf{X}$ as columns.
- The SVD of the trajectory matrix can be rewritten as:

$$\begin{aligned}\mathbf{X} &= \sum_{i=0}^{d-1} \sigma_i \mathbf{U}_i \mathbf{V}_i^T \\ &\equiv \sum_{i=0}^{d-1} \mathbf{X}_i\end{aligned}$$

- where σ_i is the i th singular value, $\mathbf{U}_i$ and $\mathbf{V}_i$ are vectors representing the i th columns of $\mathbf{U}$ and $\mathbf{V}$, respectively, $d \leq L$ is the *rank* of the trajectory matrix, and $\mathbf{X}_i = \sigma_i \mathbf{U}_i \mathbf{V}_i^T$ is the i th *elementary matrix* of $\mathbf{X}$. The collection $\{\mathbf{U}_i, \sigma_i, \mathbf{V}_i\}$ will be denoted the i th *eigentriple* of the SVD.

► Would you like to know more about the $\mathbf{U}$ matrix?

► Would you like to know more about the $\mathbf{V}$ matrix?

► Would you like to know more about the $\mathbf{\Sigma}$ matrix?

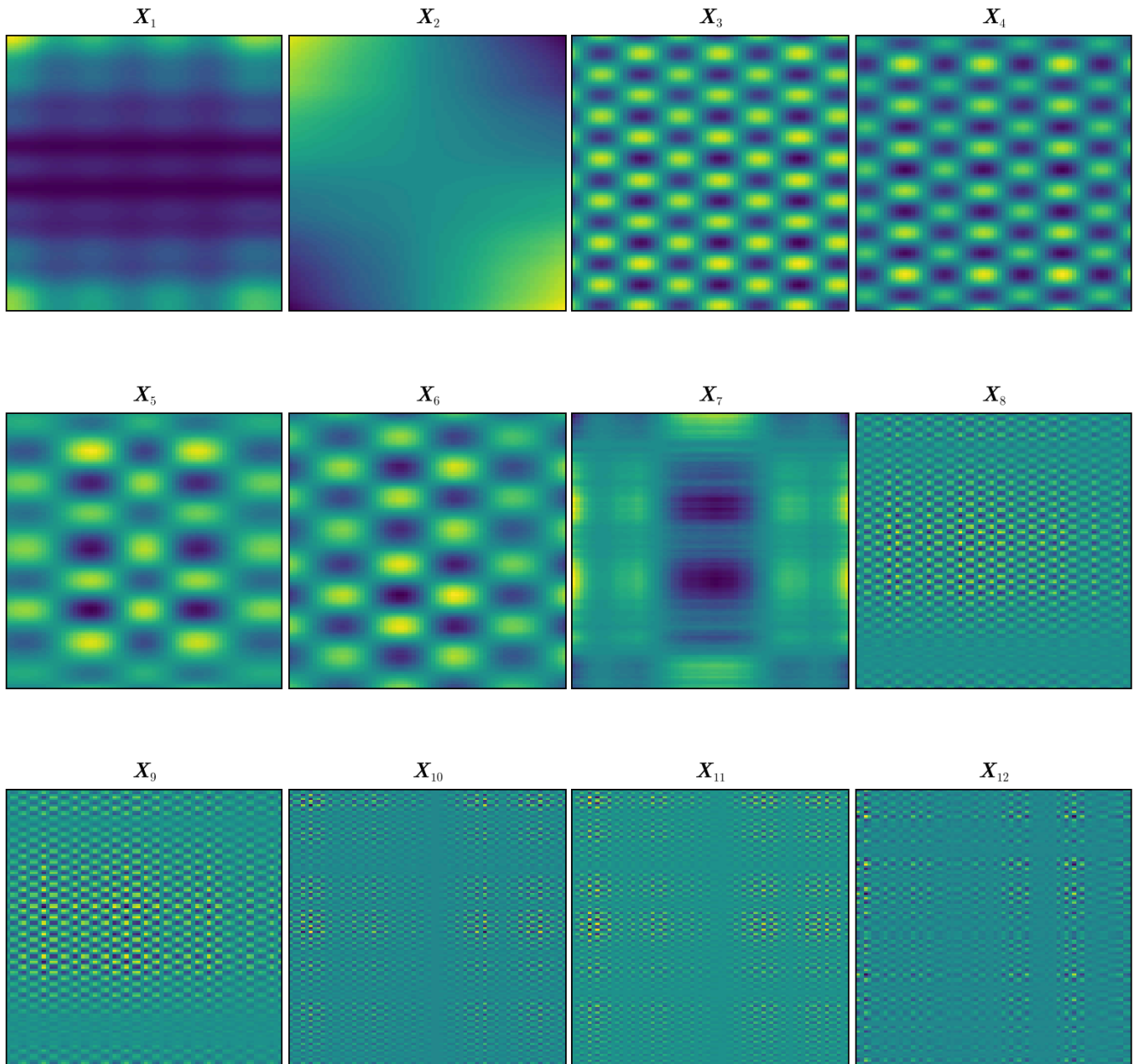
- With different words, when we decompose the trajectory matrix with an SVD; in doing so, we found two orthonormal basis sets, $\mathcal{U}$ and $\mathcal{V}$, which span the column- and row-space, respectively, of the trajectory matrix.
- From now on, we are going to focus on reconstructing the components of a time series from its elementary matrices.

```

1 begin
2     d = rank(X)    # The intrinsic dimensionality of the trajectory space.
3
4     U, Σ, V = svd(X)
5
6     X_elem = [Σ[i] * U[:, i] * (V[:, i])' for i in 1:d]
7
8     # Quick sanity check: the sum of all elementary matrices in X_elem should be equal
9     # to X, to within a
10    # *very small* tolerance:
11    if !isapprox(X, sum(X_elem), atol=1e-10)
12        println("WARNING: The sum of X's elementary matrices is not equal to X!")
13    end
14 end

```

- Let's check the first, say, 12 elementary matrices:



- From visual inspection of the $\mathbf{X}_i$ above, it is obvious that the elementary matrices lack the anti-diagonal structure of the trajectory matrix.
- Even without inspecting the $\mathbf{U}_i$ vector associated with each $\mathbf{X}_i$, or reconstructing a time series of each component, the appearance of the $\mathbf{X}_i$ hints at the nature of each component, be it trend, periodicity or noise.
 - For example, the L - and K -lagged vectors in $\mathbf{X}_1$ and $\mathbf{X}_2$ vary relatively slowly across the matrix, suggesting that $\mathbf{X}_1$ and $\mathbf{X}_2$ may be associated with the overall trend in the time series.
 - The matrices $\mathbf{X}_3$ to $\mathbf{X}_6$ show large checkerboard patterns, suggesting periodicity.
 - $\mathbf{X}_7$ may lie somewhere between periodicity and trend.
 - The matrices of $\mathbf{X}_8$ onwards (and all the way up to $\mathbf{X}_{70}$) appear to alternate quickly between a few values; these elementary matrices are likely to be associated with the noise in the original time series.

The relative contributions

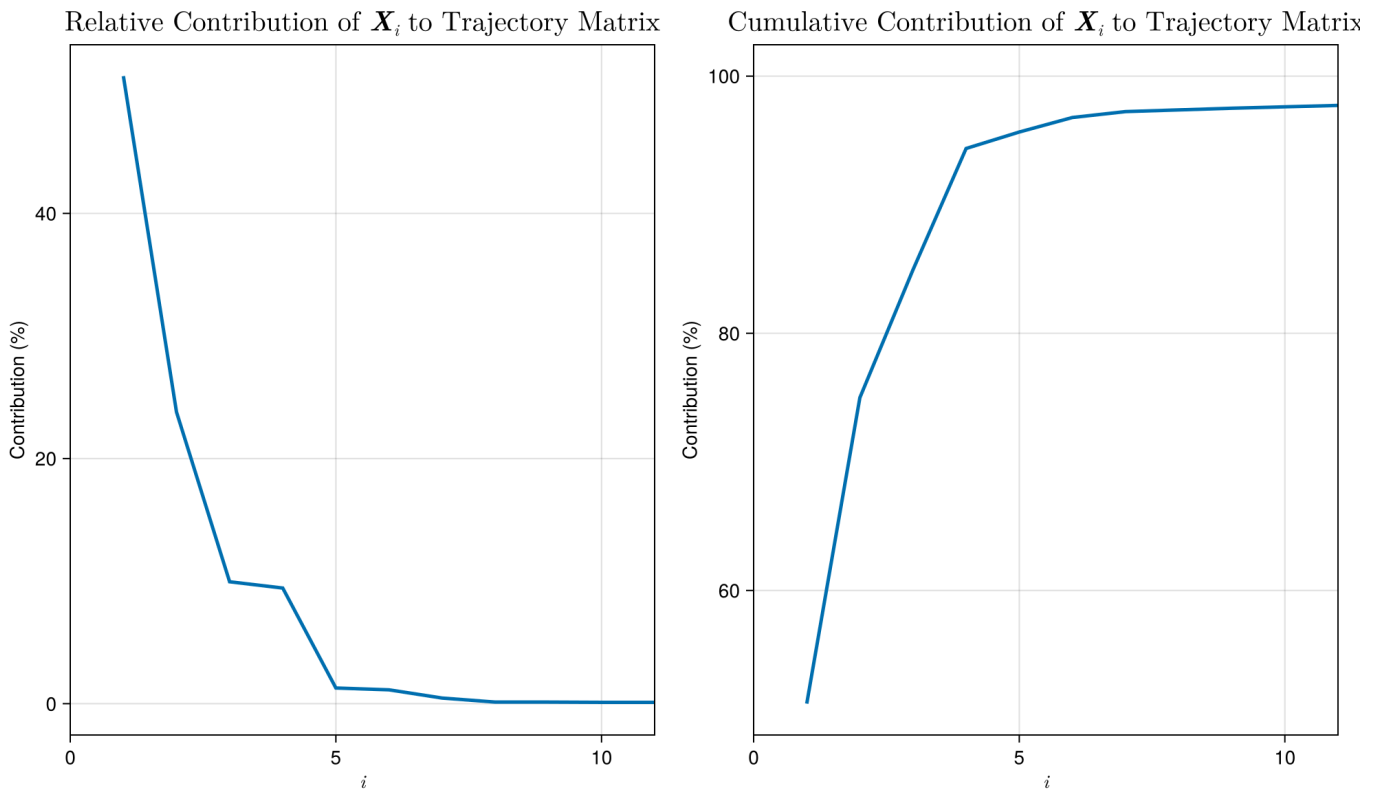
- Let's plot now the relative contributions:

$$\frac{\sigma_i^2}{\sum_{k=1}^d \sigma_k^2}$$

- and the cumulative contributions:

$$\frac{\sum_{j=1}^i \sigma_j^2}{\sum_{k=1}^d \sigma_k^2}$$

- of the first 12 elementary matrices to the trajectory matrix of the toy time series:



- The plots above depict the relative and cumulative contributions of the first 12 $\mathbf{X}_i$ in the expansion $\mathbf{X} = \sum_{i=1}^d \mathbf{X}_i$.
- The elementary matrices $\mathbf{X}_1$ and $\mathbf{X}_2$ contribute 52% and 22%, respectively, to the expansion of $\mathbf{X}$. Together, the first seven elementary matrices contribute 97%.
- Elementary matrices that make equal contributions to the expansion (that is, $\sigma_i \approx \sigma_{i+1}$) are likely to be grouped together when reconstructing the time series, and appear as "breaks" in the plot of relative contributions. For example, the "breaks" in the plot above suggest that $\mathbf{X}_3$ and $\mathbf{X}_4$, and $\mathbf{X}_5$ and $\mathbf{X}_6$, should be grouped together.

It is important to note that the elementary matrices represent an optimal (although possibly non-unique) separation of components in trajectory space: by definition, the rows and

columns of one elementary matrix are orthogonal to the rows and columns of the other elementary matrices.

- However, this separation may not (necessarily) coincide with what we would consider a useful, interpretable 'component' of the time series. In fact, there are restrictions on the types of time series components that are exactly separable under this formalism.

Reconstructing the Time Series

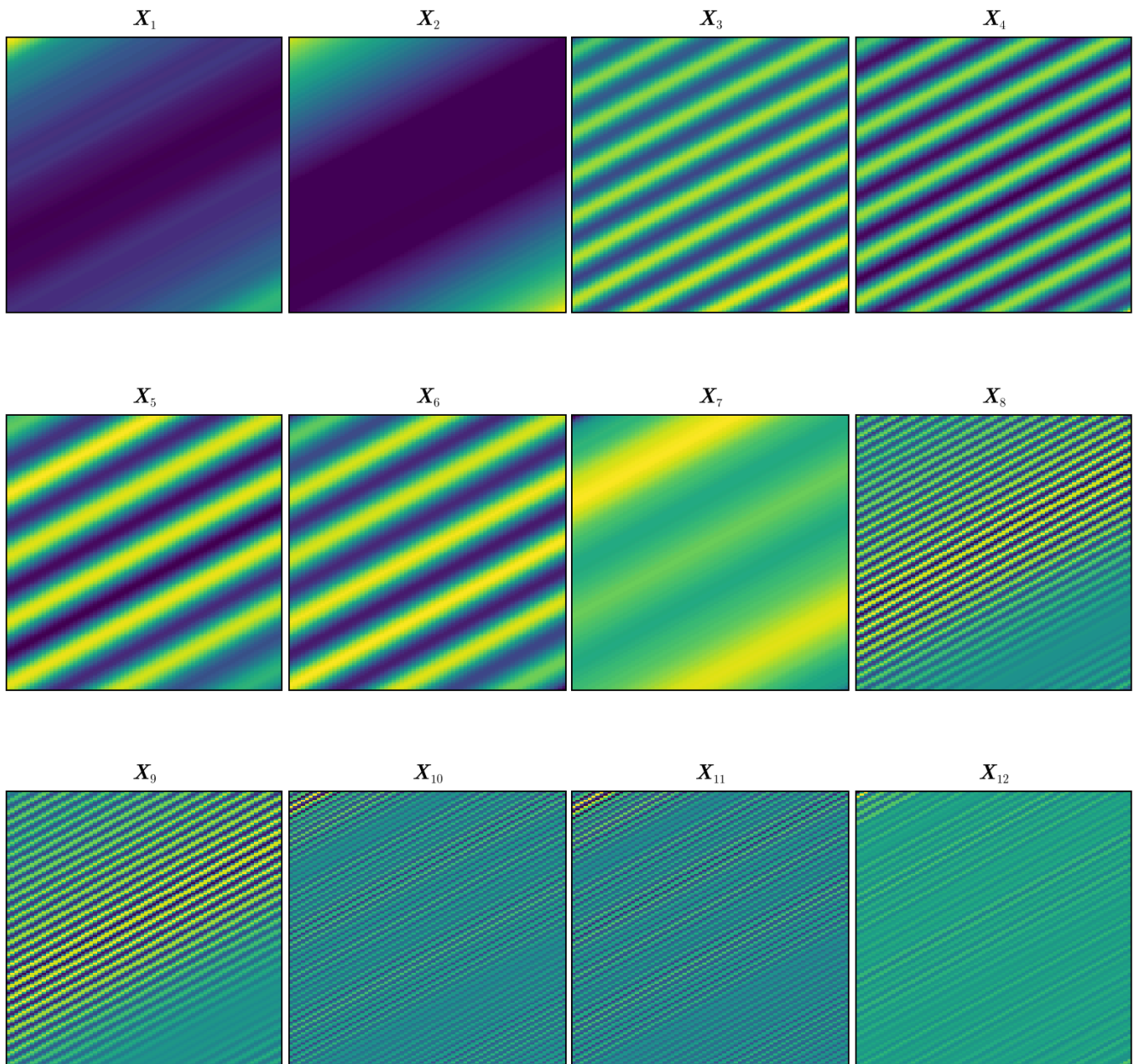
- So far, we have mapped a time series $\mathbf{F}$ to a series of L -lagged vectors, forming the *trajectory matrix* of $\mathbf{F}$.
- We then decomposed this matrix with a singular-value decomposition, and constructed a set of *elementary matrices* which comprise the trajectory matrix.
- We then gave a bit of a hand-waving explanation to classify these elementary matrices as *trend*, *periodicity* and *noise*.
- In a perfect world, all the components of a time series $\mathbf{F} = \sum_j \mathbf{F}^{(j)}$ would be separable, and we would have grouped the the resulting elementary matrices $\mathbf{X}_i$ appropriately, such that:

$$\begin{aligned}\mathbf{X} &= \sum_{k \in \mathcal{S}} \mathbf{X}_k + \sum_{l \in \mathcal{T}} \mathbf{X}_l + \dots \\ &= \sum_j \mathbf{X}^{(j)}\end{aligned}$$

- where $\mathcal{S}$ and $\mathcal{T}$ are disjoint (i.e. non-overlapping) sets of indices, and $\mathbf{X}^{(j)}$ is the trajectory matrix of the time series component $\mathbf{F}^{(j)}$.
- In this case, each $\mathbf{X}^{(j)}$ would have a Hankel structure like the original trajectory matrix, and construction of each $\mathbf{F}^{(j)}$ would be simple.
- However, in the real world, no component trajectory matrices will have equal values on their anti-diagonals. Therefore, we seek a process to transform an elementary matrix to a Hankel matrix, and then into a time series.

► Would you like to know *Hankelize* the elementary matrices?

- Once we have modified the structure of the elementary matrices we can look at them:

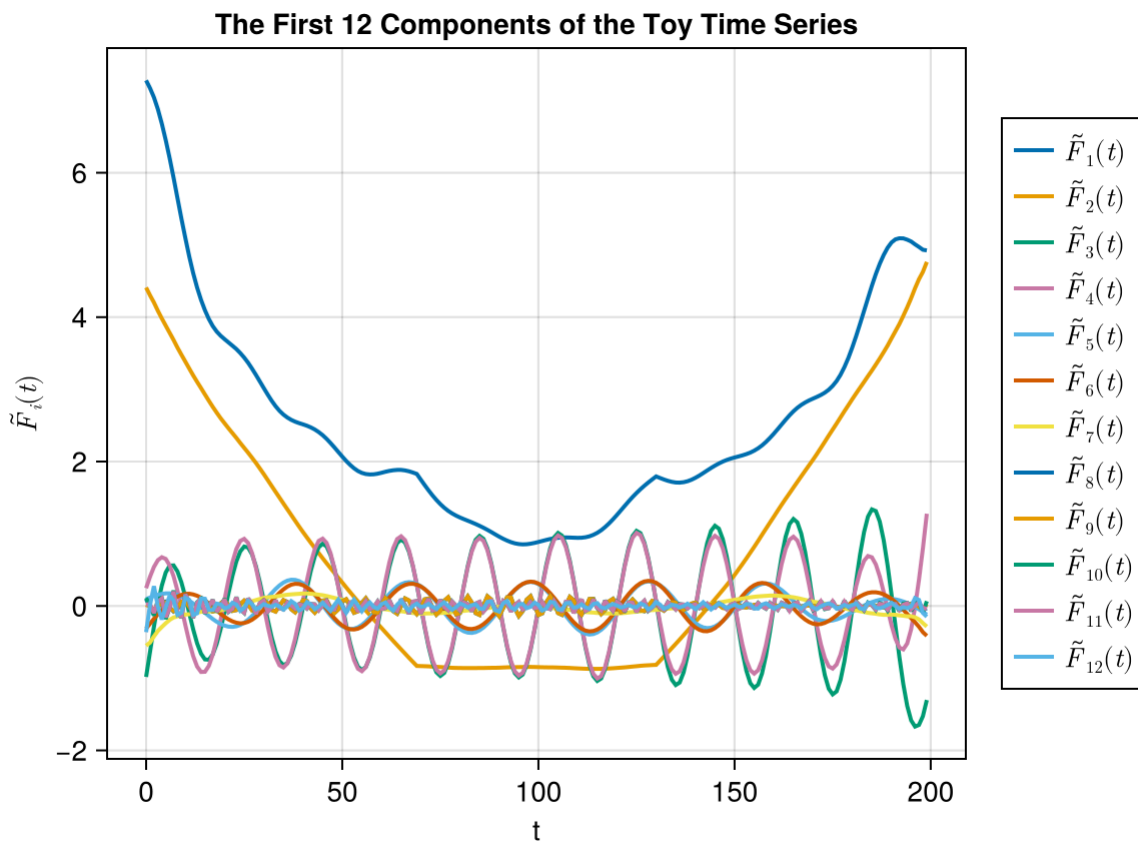


- Inspection of the Hankelised elementary matrices of the toy time series confirms our suspicions about the elementary matrices:
 - $\tilde{\mathbf{X}}_1$ and $\tilde{\mathbf{X}}_2$ vary slowly over the whole time series, and can be grouped together as the trend component.
 - $\tilde{\mathbf{X}}_3$ and $\tilde{\mathbf{X}}_4$ are both periodic, with the same frequency, and can be grouped as the first periodic component.
 - $\tilde{\mathbf{X}}_5$ and $\tilde{\mathbf{X}}_6$ are also periodic, with a different frequency to $\tilde{\mathbf{X}}_3$ and $\tilde{\mathbf{X}}_4$, and will be grouped as the second periodic component.
 - $\tilde{\mathbf{X}}_7$, lacking obvious periodicity, will be grouped with the trend components.
- We'll lump together all components from $\tilde{\mathbf{X}}_8$ and beyond as noise.
- To summarise:

$$\begin{aligned}
\tilde{\mathbf{X}}^{(\text{trend})} &= \tilde{\mathbf{X}}_1 + \tilde{\mathbf{X}}_2 + \tilde{\mathbf{X}}_7 & \Rightarrow \tilde{\mathbf{F}}^{(\text{trend})} &= \tilde{\mathbf{F}}_1 + \tilde{\mathbf{F}}_2 + \tilde{\mathbf{F}}_7 \\
\tilde{\mathbf{X}}^{(\text{periodic 1})} &= \tilde{\mathbf{X}}_3 + \tilde{\mathbf{X}}_4 & \Rightarrow \tilde{\mathbf{F}}^{(\text{periodic 1})} &= \tilde{\mathbf{F}}_3 + \tilde{\mathbf{F}}_4 \\
\tilde{\mathbf{X}}^{(\text{periodic 2})} &= \tilde{\mathbf{X}}_5 + \tilde{\mathbf{X}}_6 & \Rightarrow \tilde{\mathbf{F}}^{(\text{periodic 2})} &= \tilde{\mathbf{F}}_5 + \tilde{\mathbf{F}}_6 \\
\tilde{\mathbf{X}}^{(\text{noise})} &= \tilde{\mathbf{X}}_8 + \dots + \tilde{\mathbf{X}}_{70} & \Rightarrow \tilde{\mathbf{F}}^{(\text{noise})} &= \tilde{\mathbf{F}}_8 + \dots + \tilde{\mathbf{F}}_{70}
\end{aligned}$$

- While we have defined the time series component grouping in terms of Hankelised elementary matrices, we will no longer calculate the full Hankel matrix $\tilde{\mathbf{X}}_i$, and instead calculate $\tilde{\mathbf{F}}_i$ directly from $\mathbf{X}_i$.

- Let's go ahead and construct the first 12 elementary components, $\tilde{\mathbf{F}}_i$, for the toy time series.

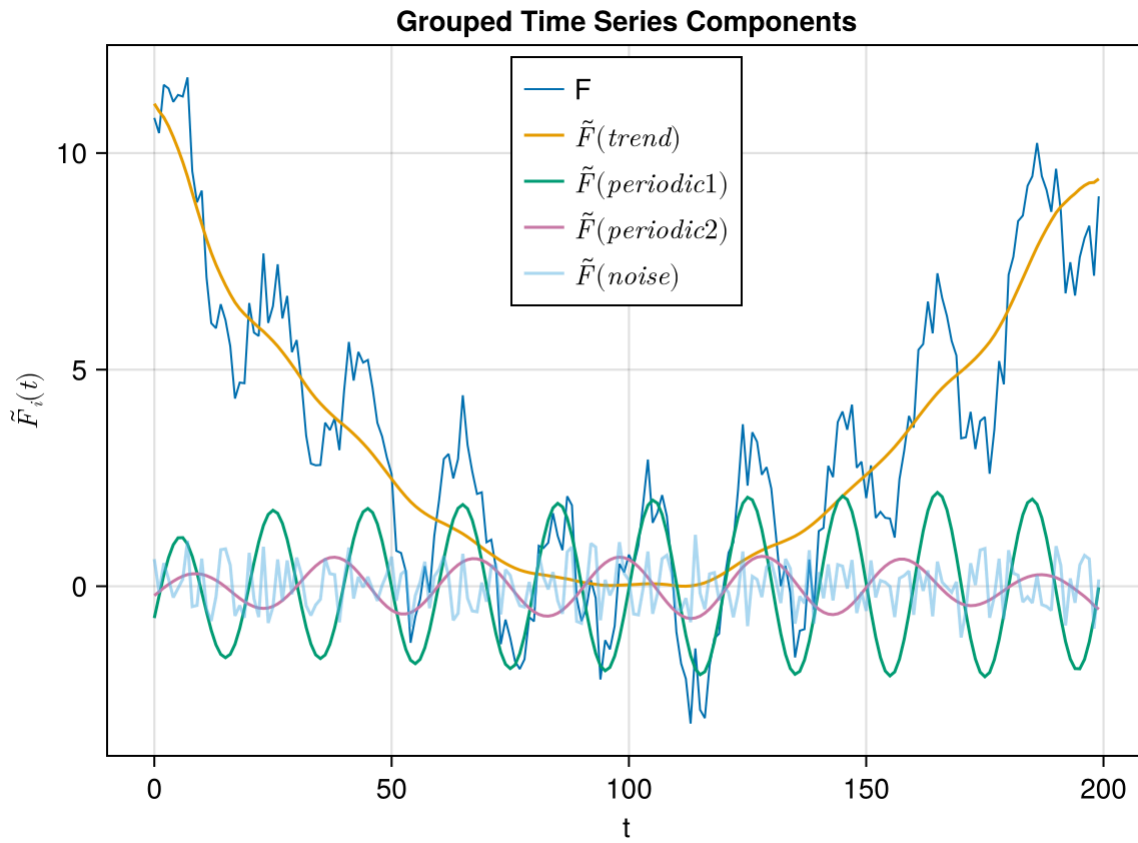


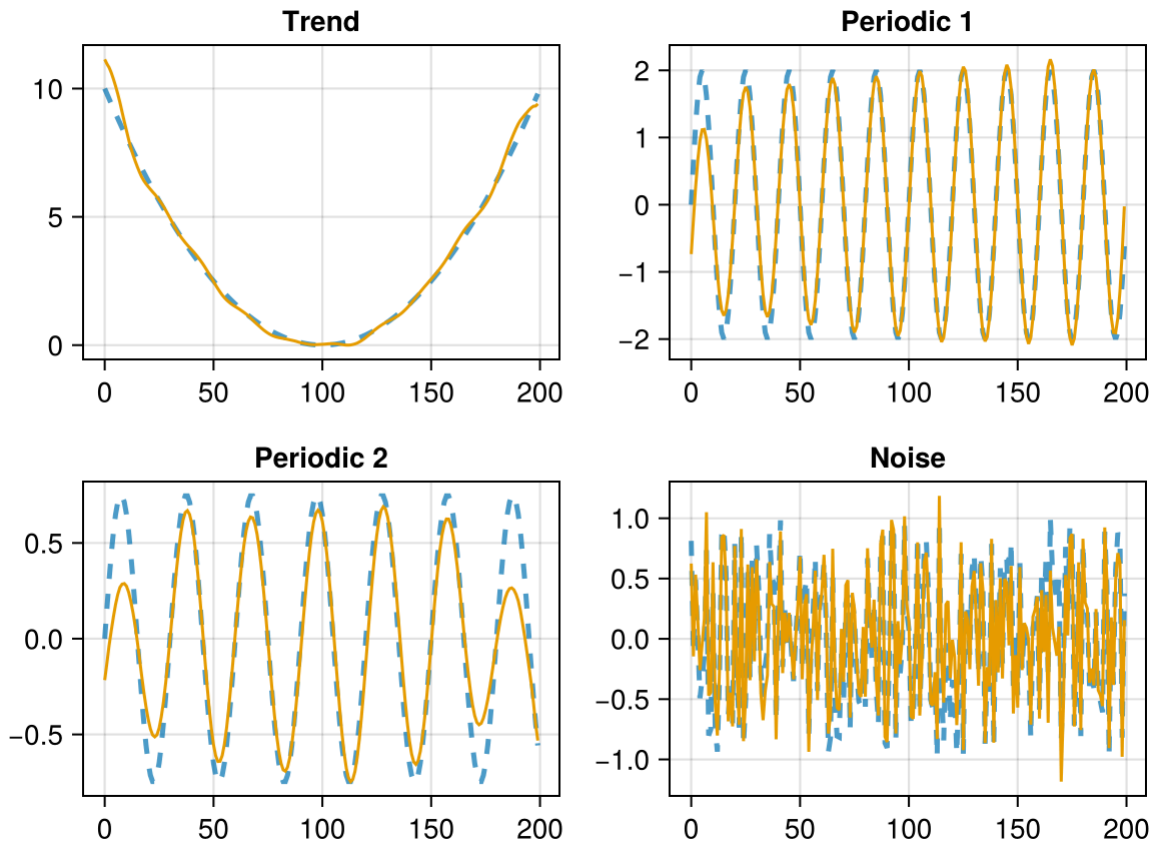
- As mentioned earlier, the elementary components separated in the time series' trajectory space may not coincide with a single, interpretable component in the time series.
 - For example, $\tilde{\mathbf{F}}_1$ and $\tilde{\mathbf{F}}_2$ both look vaguely like the trend—are they *really* separate components? Similarly, $\tilde{\mathbf{F}}_3$ and $\tilde{\mathbf{F}}_4$ are almost identical, except near the boundaries of the time series.
- We'll introduce a way to quantify which $\tilde{\mathbf{F}}_i$ should be grouped together, but for a moment, let's follow our instincts and apply our earlier grouping for $\tilde{\mathbf{F}}^{(\text{trend})}$, $\tilde{\mathbf{F}}^{(\text{periodic 1})}$, $\tilde{\mathbf{F}}^{(\text{periodic 2})}$ and $\tilde{\mathbf{F}}^{(\text{noise})}$, and see how the SSA-separated components compare with the original components that compose the toy time series:

```

1 begin
2     # Assemble the grouped components of the time series.
3     F_trend = X_to_TS(sum(X_elem[[1, 2, 7], :], dims=1)[1])
4     F_periodic1 = X_to_TS(sum(X_elem[[3, 4], :], dims=1)[1])
5     F_periodic2 = X_to_TS(sum(X_elem[[5, 6], :], dims=1)[1])
6     F_noise = X_to_TS(sum(X_elem[8:end, :], dims=1)[1]);
7 end;

```





- After grouping the elementary components together, it looks like SSA has done a great job separating the original components of the toy time series—especially the two periodic components with differing frequencies and amplitudes.
 - However, the separation isn't perfect: all the components deteriorate near the boundaries, especially the trend and second periodic component.
 - This is common in SSA, and arises from the fact that, under the SSA formalism, most types of series (i.e. polynomial, sine, exponential, etc.) are not exactly separable. Therefore, our attempt to recover the *exact* parabolic trend and periodic components from the toy series was always doomed to fail.
- However, that does not mean components of a time series cannot be *approximately separable*, as we witnessed above (there is also the concept of *asymptotic separability* when the length of the time series approaches infinity).

Time Series Component Separation and Grouping

- So far, we have grouped the eigentriples/components of the toy time series together by visual inspection; that is, we decided which components belonged together by their appearance.
- This is fine for a short and simple time series, however, for longer and more complicated time series, we seek a method that quantifies whether a reconstructed component $\tilde{\mathbf{F}}_i$ can be considered separate from another component $\tilde{\mathbf{F}}_j$, so we don't need to make grouping decisions by visually inspecting each $\tilde{\mathbf{F}}_i$.
- For two reconstructed time series, $\tilde{\mathbf{F}}_i$ and $\tilde{\mathbf{F}}_j$, of length N , and a window length L , we define the *weighted inner product*, $(\tilde{\mathbf{F}}_i, \tilde{\mathbf{F}}_j)_w$ as:

$$(\tilde{\mathbf{F}}_i, \tilde{\mathbf{F}}_j)_w = \sum_{k=0}^{N-1} w_k \tilde{f}_{i,k} \tilde{f}_{j,k}$$

- where $\tilde{f}_{i,k}$ and $\tilde{f}_{j,k}$ are the k th values of $\tilde{\mathbf{F}}_i$ and $\tilde{\mathbf{F}}_j$, respectively, and w_k is given by

$$w_k = \begin{cases} k+1 & 0 \leq k \leq L-1 \\ L & L \leq k \leq K-1 \\ N-k & K \leq k \leq N-1 \end{cases}$$

- remembering that $K = N - L + 1$.
- The weight w_k simply reflects the number of times $\tilde{f}_{i,k}$ and $\tilde{f}_{j,k}$ appear in the Hankelised matrices $\tilde{\mathbf{X}}_i$ and $\tilde{\mathbf{X}}_j$, from which the time series $\tilde{\mathbf{F}}_i$ and $\tilde{\mathbf{F}}_j$ have been obtained.
- Put simply, if $(\tilde{\mathbf{F}}_i, \tilde{\mathbf{F}}_j)_w = 0$, $\tilde{\mathbf{F}}_i$ and $\tilde{\mathbf{F}}_j$ are *w-orthogonal* and the time series components are separable.
 - Of course, total w-orthogonality does not occur in real life, so instead we define a $d \times d$ *weighted correlation* matrix, $\mathbf{W}_{\text{corr}}$, which measures the deviation of the components $\tilde{\mathbf{F}}_i$ and $\tilde{\mathbf{F}}_j$ from w-orthogonality.
- The elements of $\mathbf{W}_{\text{corr}}$ are given by:

$$W_{i,j} = \frac{(\tilde{\mathbf{F}}_i, \tilde{\mathbf{F}}_j)_w}{\|\tilde{\mathbf{F}}_i\|_w \|\tilde{\mathbf{F}}_j\|_w}$$

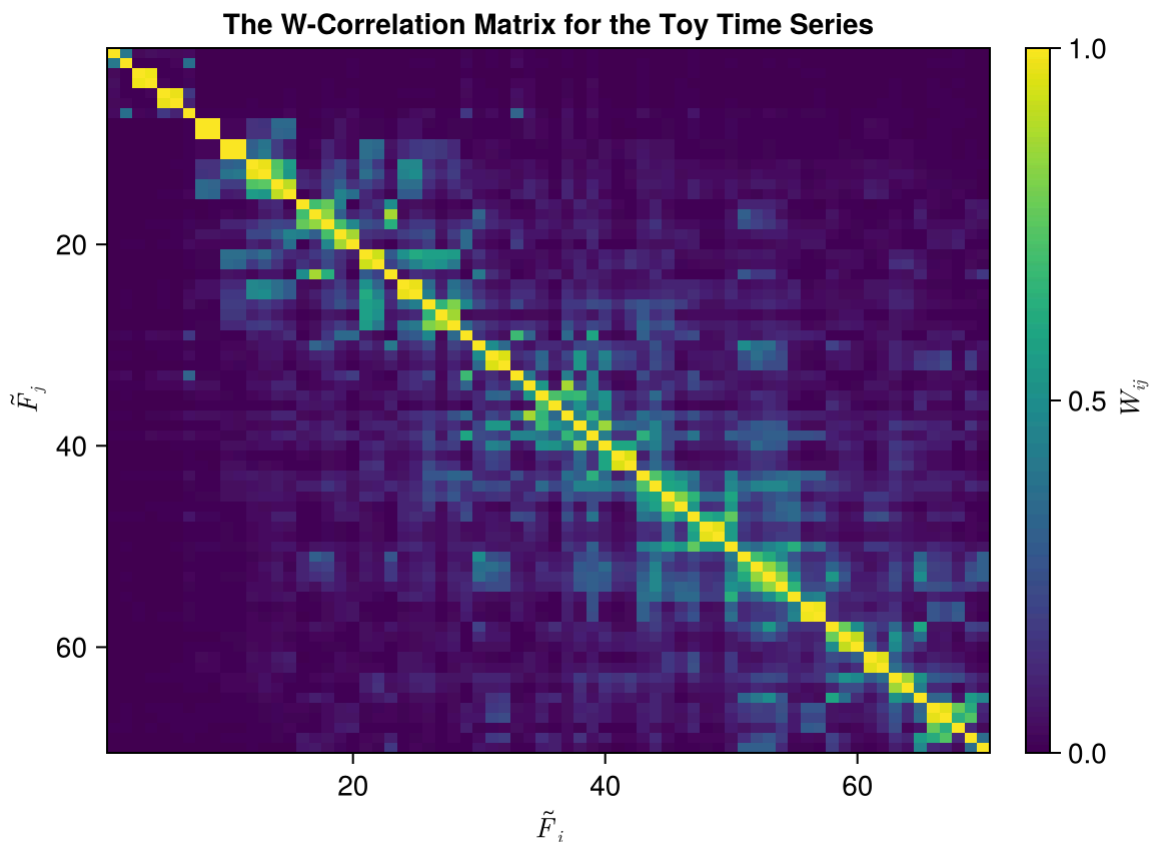
- where $\|\tilde{\mathbf{F}}_k\|_w = \sqrt{(\tilde{\mathbf{F}}_k, \tilde{\mathbf{F}}_k)_w}$ for $k = i, j$.
- The interpretation of $W_{i,j}$ is straightforward: if $\tilde{\mathbf{F}}_i$ and $\tilde{\mathbf{F}}_j$ are arbitrarily close together (but not identical), then $(\tilde{\mathbf{F}}_i, \tilde{\mathbf{F}}_j)_w \rightarrow \|\tilde{\mathbf{F}}_i\|_w \|\tilde{\mathbf{F}}_j\|_w$ and therefore $W_{i,j} \rightarrow 1$.
 - Of course, if $\tilde{\mathbf{F}}_i$ and $\tilde{\mathbf{F}}_j$ are w-orthogonal, then $W_{i,j} = 0$.
- Moderate values of $W_{i,j}$ between 0 and 1, say $W_{i,j} \geq 0.3$, indicate components that may need to be grouped together.
- Let's now construct the w-correlation matrix for the toy time series:

```

1 begin
2   w = vcat(
3     collect(1:L),
4     repeat([L], K - L - 1),
5     collect(L:-1:1)
6   ) |> x -> convert(Vector{Float64}, x)
7
8   F_elem = [X_to_TS(X_elem[i]) for i in 1:d]
9
10  F_wnorms_sq = [w' * (F_elem[i] .^ 2) for i in 1:d]
11  F_wnorms = F_wnorms_sq .^ -0.5
12
13  Wcorr = Matrix{Float64}(I, d, d)
14
15  for i in 1:d
16    for j in (i+1):d
17      dot_product = w' * (F_elem[i] .* F_elem[j])
18
19      Wcorr[i, j] = abs(dot_product * F_wnorms[i] * F_wnorms[j])
20      Wcorr[j, i] = Wcorr[i, j]
21    end
22  end
23 end

```

- Plot the w-correlation matrix.



- The structure of $\mathbf{W}_{\text{corr}}$ shows a lot of correlation between the time series components, particularly in the range $8 \leq i, j \leq 70$. As these were the components we classified as

belonging to the noise in the time series, it is no surprise that there are non-negligible correlations between all of them; this is a natural result of the noise having no underlying structural component that can be further separated.

- It is important to note that $\mathbf{W}_{\text{corr}}$ is roughly split into two 'blocks': $1 \leq i, j \leq 7$, and $8 \leq i, j \leq 70$. This corresponds to two main groupings: a smoothed time series (i.e. the trend plus the two periodic components), and the residual noise. Zooming into the first seven components in $\mathbf{W}_{\text{corr}}$:



- The initial appearance-based groupings we made for the first six components are supported by the corresponding w-correlation values. $\tilde{F}_1$ and $\tilde{F}_2$ have $W_{1,2} = 0.40$, suggesting they should be paired. $\tilde{F}_2$ and $\tilde{F}_7$ also have $W_{2,7} = 0.39$, suggesting $\tilde{F}_7$ should also be grouped with $\tilde{F}_1$ and $\tilde{F}_2$ as a trend component. However, $\tilde{F}_7$ also has a slight w-correlation with $\tilde{F}_5$ and $\tilde{F}_6$, but since $\tilde{F}_5$ and $\tilde{F}_6$ have no w-correlation with $\tilde{F}_1$ and $\tilde{F}_2$, we choose to keep $\tilde{F}_7$ with $\tilde{F}_1$ and $\tilde{F}_2$.
- Our prior groupings of $\tilde{F}^{(\text{periodic } 1)} = \tilde{F}_3 + \tilde{F}_4$ and $\tilde{F}^{(\text{periodic } 2)} = \tilde{F}_5 + \tilde{F}_6$ are clearly justified by the w-correlation matrix, with $W_{3,4} = 0.98$ and $W_{5,6} = 0.98$.

A julia module for SSA

- It is time to collect the SSA code into a handy module, imaginatively named *SSAJL*, which will form the basis for the rest of this notebook. Each instance of the class will contain the decomposition of a time series for some window length L , and provide useful methods to analyse, plot and reconstruct the time series.
- To summarise the SSA algorithm:
 1. For a time series $F = (f_0, f_1, \dots, f_{N-1})$, and a window length L , form the trajectory matrix $\mathbf{X}$, with columns given by the vectors $(f_i, \dots, f_{L+i-1})^T$, $0 \leq i \leq N - L$.
 2. Decompose $\mathbf{X}$ with the singular value decomposition, $\mathbf{X} = \sum_{i=0}^{d-1} \sigma_i U_i V_i^T$.
 3. Construct the d elementary matrices $\mathbf{X}_i = \sigma_i U_i V_i^T$.
 4. Diagonally average the $\mathbf{X}_i$ to form the elementary time series components $\tilde{F}_i$, such that $F = \sum_{i=0}^{d-1} \tilde{F}_i$.
 5. Calculate and store the weighted correlation matrix, $\mathbf{W}_{\text{corr}}$, for the $\tilde{F}_i$.
- The task of grouping and classifying the elementary components $\tilde{F}_i$ is left to the user.

Main.var"workspace#3".SSAJL

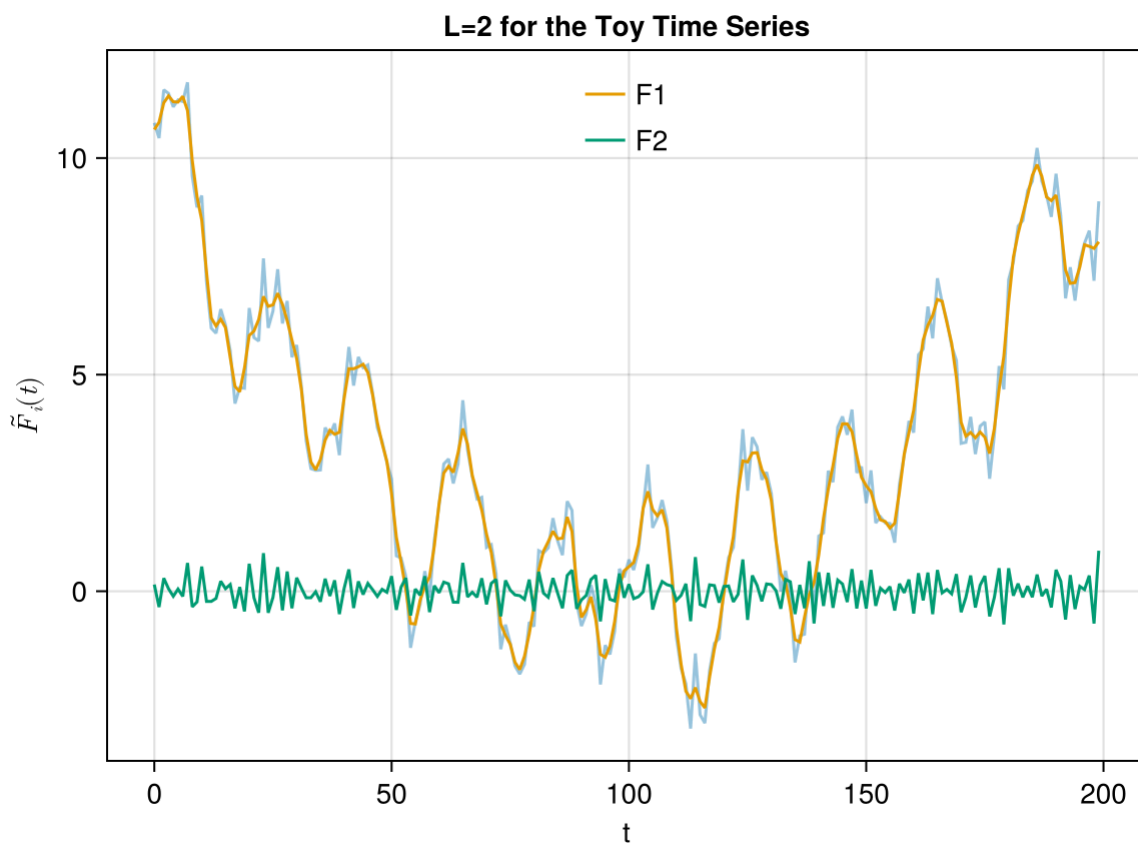
The Window Length

- We have now established the machinery to easily investigate the effect of the window length parameter, L , on the decomposition of our toy time series.

$L = 2$

- A window length of 2 may seem like a useless choice, but it's a good place to start and watch the time series get decomposed into more and more components.

```
1 begin
2     F_ssa_L2 = SSAJL.SSA(F, 2)
3
4     cpmm = SSAJL.components_to_matrix(F_ssa_L2)
5     orts = F_ssa_L2.orig_TS
6 end;
```



- For $L = 2$ we can only expect two elementary components to be returned. Even for such a small window length, the SSA algorithm has started to separate the high-frequency noise from the series, giving us a somewhat-denoised version of the original series in the component $\tilde{F}_1$.

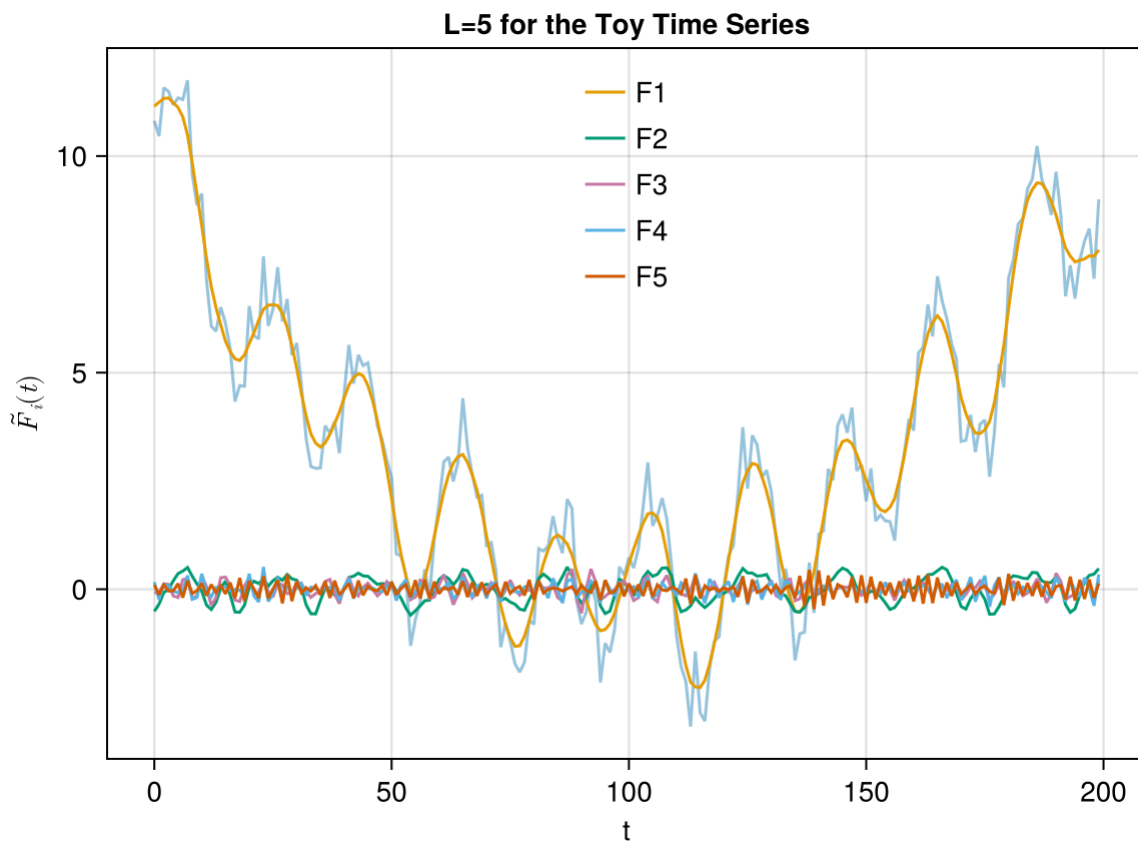
$L = 5$

- Let's go up to a window length of 5, and see what happens to the elementary components:

```

1 begin
2   F_ssa_L5 = SSAJL.SSA(F, 5)
3
4   cpmm8 = SSAJL.components_to_matrix(F_ssa_L5)
5   orts8 = F_ssa_L5.orig_TS
6 end;

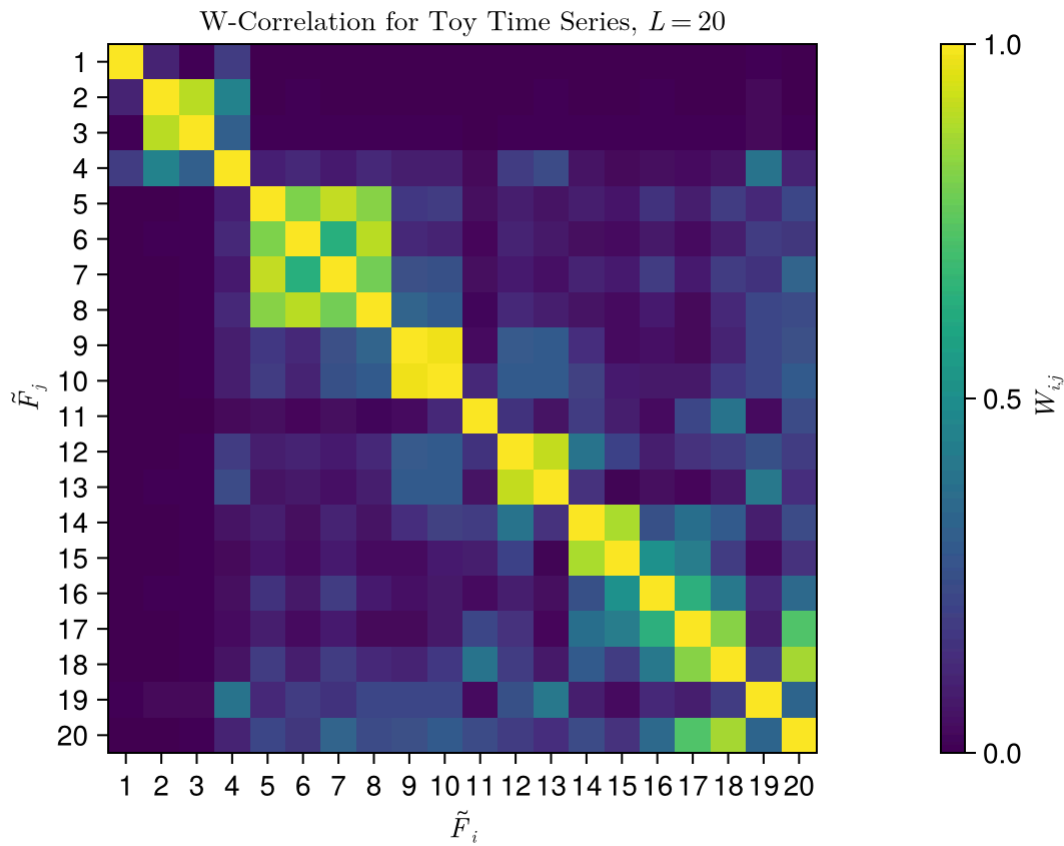
```



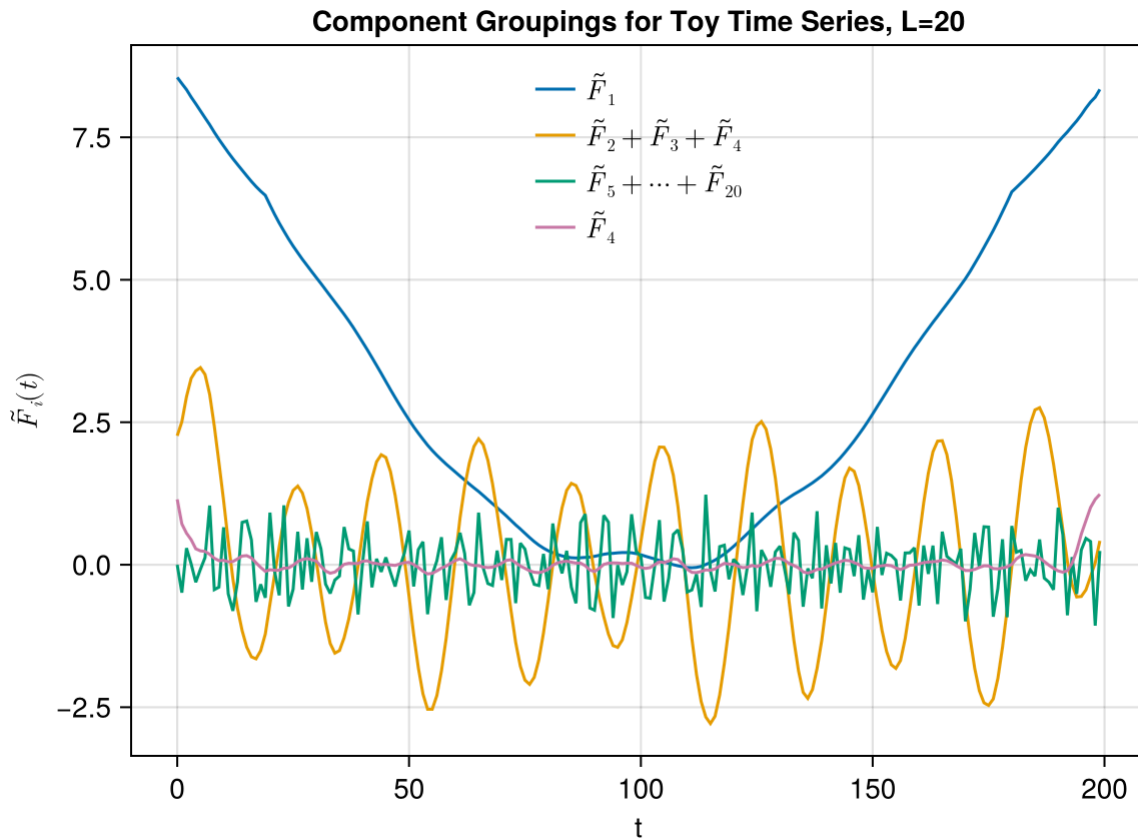
- We see that $\tilde{F}_1$ is now a well-and-truly denoised version of the original series. $\tilde{F}_2$ is a poorly resolved periodic component, while $\tilde{F}_3$ to $\tilde{F}_5$ are just noise.

$L = 20$

- Let's quadruple the window length, and instead of inspecting elementary components, we'll look at the resulting w-correlation matrix and make some grouping decisions first.



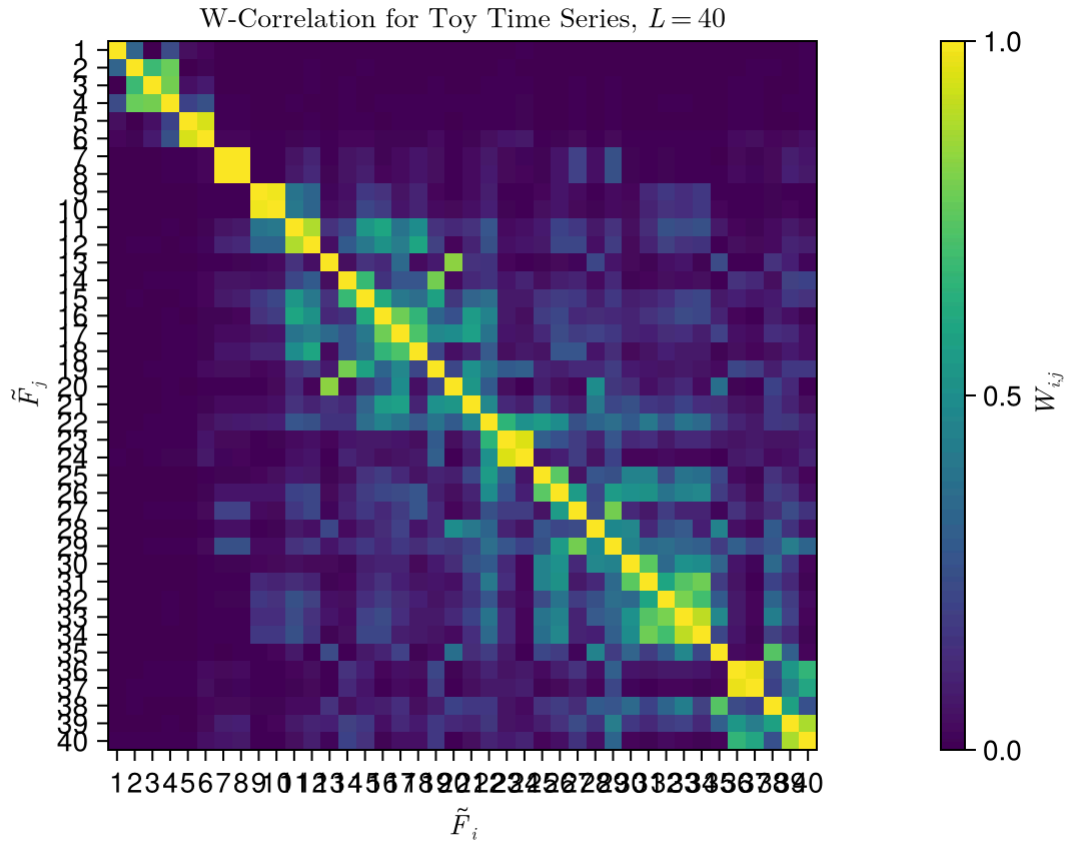
- The w-correlation matrix for $L = 20$ is split (roughly) into two blocks: $\tilde{F}_1$ to $\tilde{F}_4$, and $\tilde{F}_5$ to $\tilde{F}_{20}$.
- Within those blocks, the size of the $W_{i,j}$ values suggest that we need to group $\tilde{F}_2$, $\tilde{F}_3$ and $\tilde{F}_4$, and group all $\tilde{F}_5, \dots, \tilde{F}_{20}$.
- This grouping is certainly not ideal, as $\tilde{F}_4$ has non-negligible w-correlation with components in the second block. We'll plot our chosen component groupings, along with $\tilde{F}_4$ on its own, and see if we're justified in our choice of grouping:



- For $L = 20$ we begin to see the trend and periodic components start to take shape. The single component $\tilde{F}_1$ looks like the parabolic trend, and the group $\tilde{F}_2 + \tilde{F}_3 + \tilde{F}_4$ is a very handsome periodicity, corresponding to the sum of the two periodic components in the original definition of the toy time series.
 - The component $\tilde{F}_4$ is indeed troublesome, looking like it contributes to both noise *and* periodicity. This suggests we need to increase the window length and see if we get an improved separation of noise and periodicity.

$L = 40$

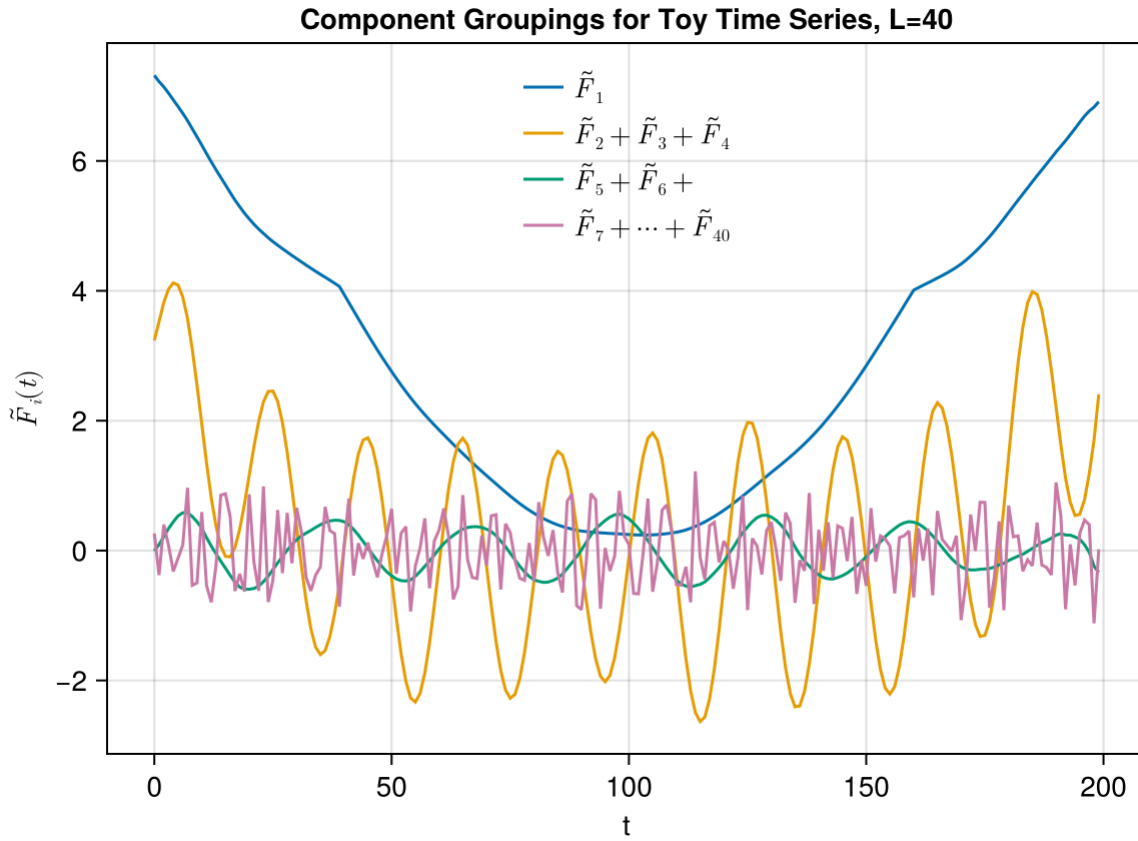
- Once again, we'll double the window length and inspect the w-correlation matrix first:



- The w-correlation matrix for $L = 40$ retains the two-block structure, with $\tilde{F}_1, \dots, \tilde{F}_6$ in the first block, and $\tilde{F}_7, \dots, \tilde{F}_{40}$ in the second. Let us group the components as follows:

$$\begin{aligned}\tilde{F}^{(1)} &= \tilde{F}_1 \\ \tilde{F}^{(2)} &= \tilde{F}_2 + \tilde{F}_3 + \tilde{F}_4 \\ \tilde{F}^{(3)} &= \tilde{F}_5 + \tilde{F}_6 \\ \tilde{F}^{(4)} &= \tilde{F}_7 + \dots + \tilde{F}_{40}\end{aligned}$$

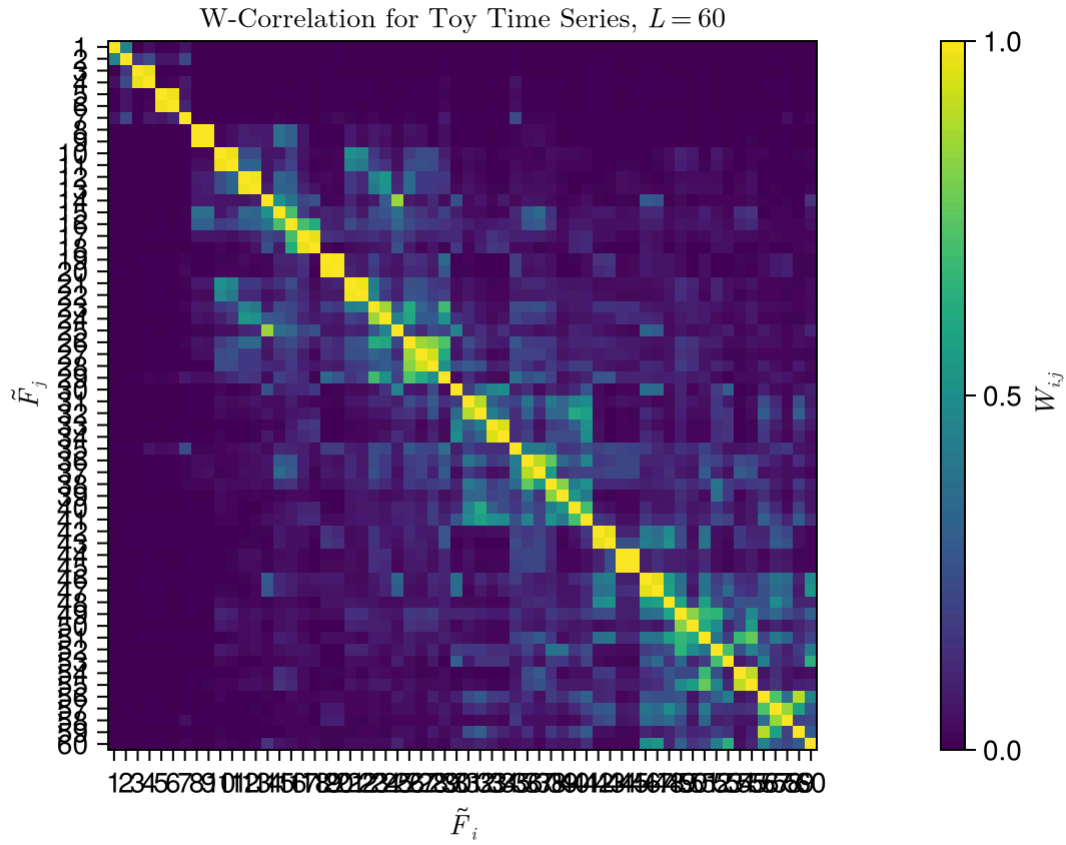
- Once again, it can be argued that this grouping is not ideal, given the non-negligible w-correlation between, for example, $\tilde{F}_1$ and $\tilde{F}_2$.



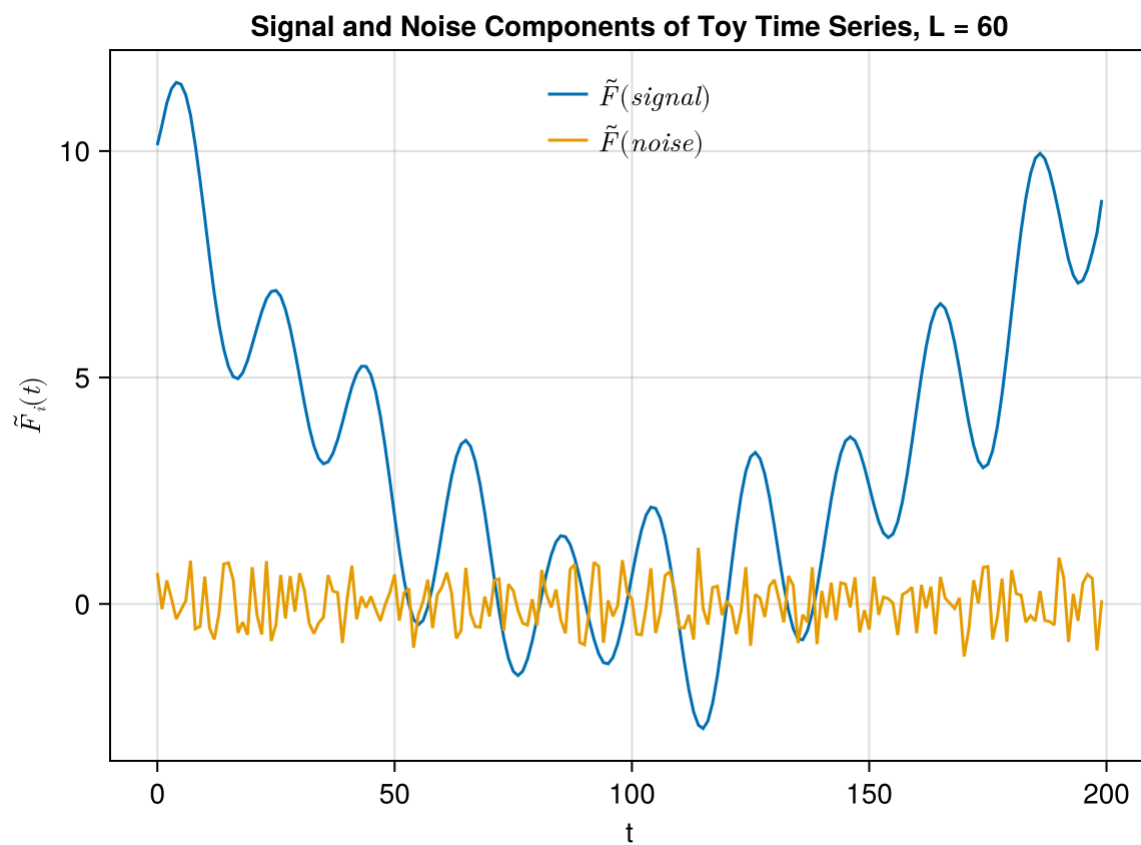
- Interestingly, the trend component $\tilde{F}^{(1)}$ has started to deteriorate at $L = 40$, with notable 'kinks' in the time series.
- The periodicity in the toy series is now separated into two periodic components of differing amplitudes and frequencies, however with significant deteriorations near the beginning and end of the time series.
- At this stage our window length is 20% of the length of the time series, but the poor quality of the separated components suggests that we still need to increase the window size.

$L = 60$

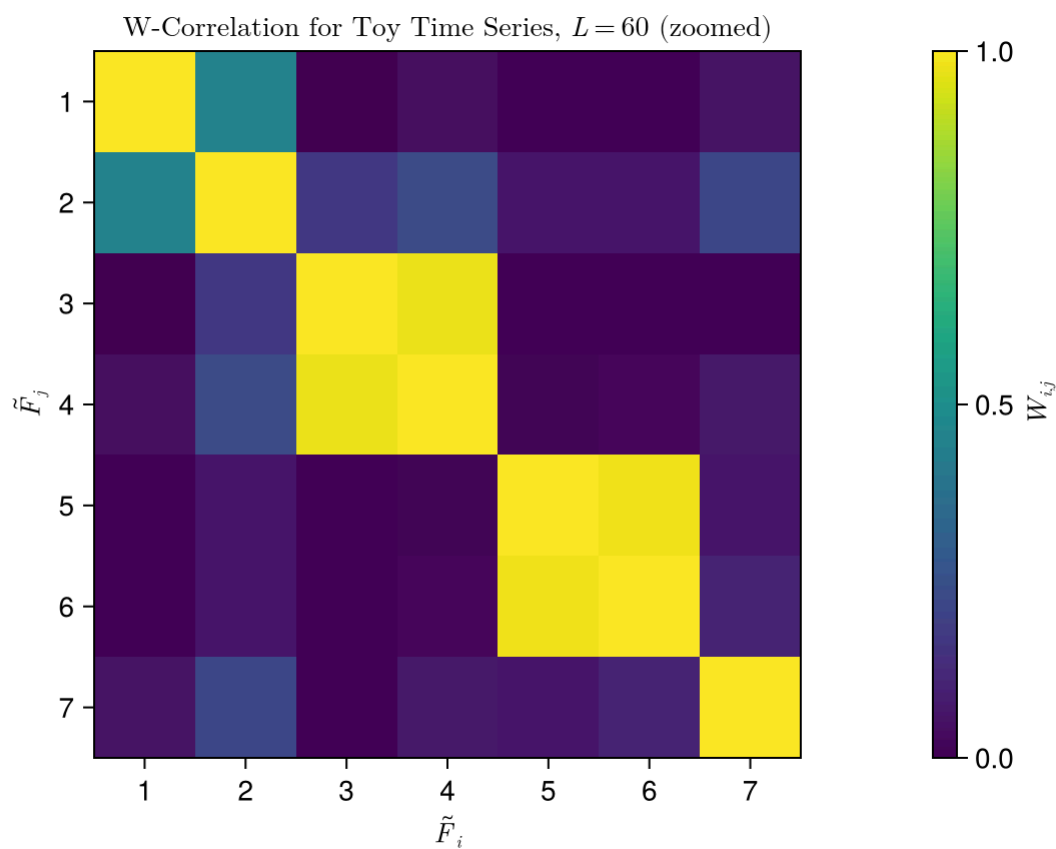
- We are now approaching our first choice of $L = 70$, so it's worth investigating how the decomposition converges to our original results. Inspect the w-correlation matrix first:



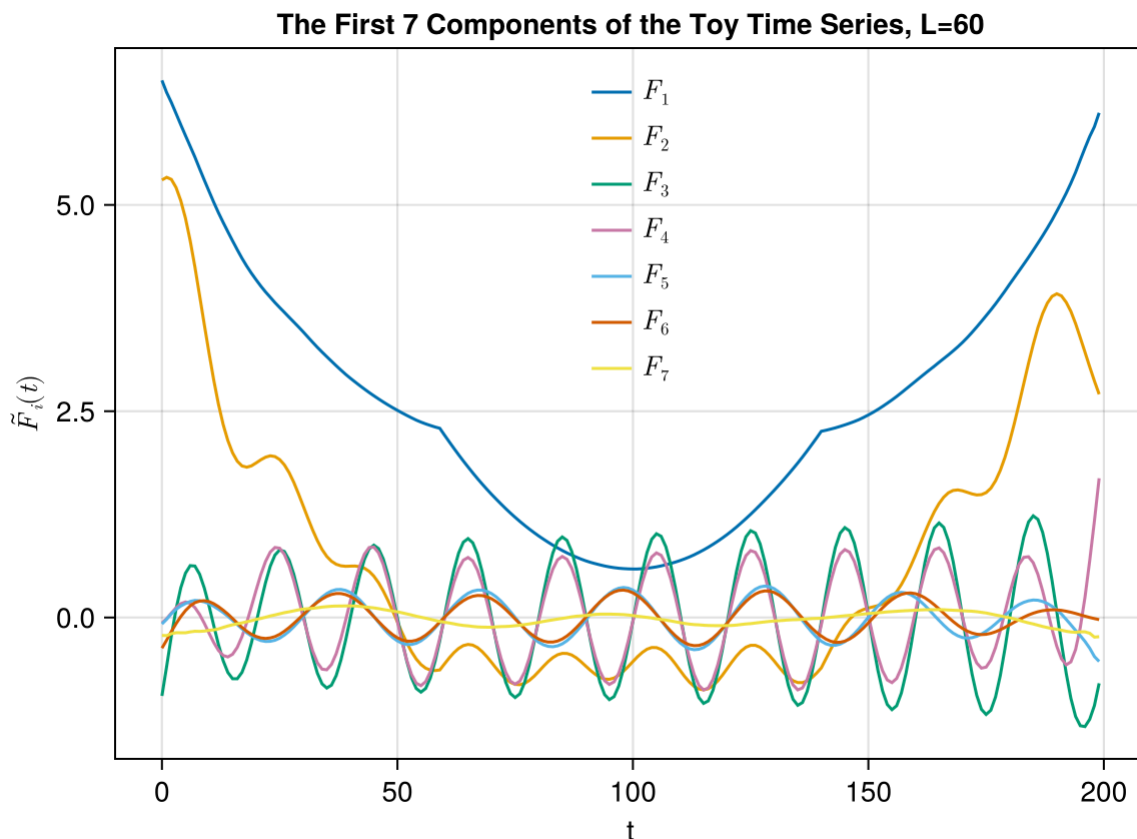
- As with the original $L = 70$ result, the w-correlation matrix is now composed of two separate blocks: $\tilde{F}_1$ to $\tilde{F}_7$, and $\tilde{F}_8$ to $\tilde{F}_{60}$.
- From experience now, it is clear that $\tilde{F}^{(\text{signal})} = \sum_{i=1}^7 \tilde{F}_i$ will be the combined trend and periodic components ('signal'), and $\tilde{F}^{(\text{noise})} = \sum_{i=8}^{60} \tilde{F}_i$ will be the noise:



- It is also of interest viewing the w-correlation matrix for components 1–7:



- In order to understand why there is non-negligible w-correlation between most of the first seven components, it'll be prudent to plot all of them at once:



- It is clear from the plot above that the trend component $\tilde{F}_2$ contains oscillations that are the same frequency as the periodic components $\tilde{F}_3$ and $\tilde{F}_4$, resulting in substantial w-correlations.
- As we've already witnessed, increasing the window length to $L = 70$ is enough to remove these oscillations almost entirely.
 - It is certainly possible to increase the window length further—up to $L = N/2 = 100$ in this case—however, beyond $L = 70$, the time series components generated by the leading seven eigentriples are (relatively) insensitive to the window length.
- In SSA, there are no hard and fast rules to setting the perfect window length, beyond $2 \leq L \leq N/2$. However, longer window lengths (up to 30–45% of the length of the time series) are sometimes required to adequately separate underlying periodicities from the overall trend. Some trial and error is needed, but it is often easy to start at a 'large-enough' window length, and work from there.

Reference & Material

Material and papers related to the topics discussed in this lecture.

- [Hassani \(2007\) - "Singular Spectrum Analysis: Methodology and Comparison"](#)

- [Rico et al. \(2025\) - "Singular spectrum analysis of Fermi-LAT blazar light curves: A systematic search for periodicity and trends in the time domain"](#)

Further Material

Papers for examining more closely some of the discussed topics.

- [Deng \(2014\) - "Time Series Decomposition Using Singular Spectrum Analysis "](#)
- [Golyandina et al. \(2001\) - "Analysis of Time Series Structure: SSA and Related Techniques"](#)

Credits

This notebook is obtained and further elaborated from:

<https://www.kaggle.com/code/jdarcy/introducing-ssa-for-time-series-decomposition>.

Course Flow

	Previous lecture	Next lecture	Course Summary
notebook	Lecture about non-parametric periodograms	Science case about motion sensor data	Course Summary
html	Lecture about non-parametric periodograms	Science case about motion sensor data	Course Summary

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